Huffman Coding

Topic: Greedy File: g3.pdf

PROBLEM

Build a prefix-free binary code with minimum weighted path length for symbols with known frequencies.

GREEDY CHOICE

Put all symbol frequencies in a min-priority queue. Repeatedly remove the two smallest nodes and combine them into a parent with summed frequency.

ALGORITHM

Insert each symbol as a leaf. While more than one node remains, merge the two minimum-frequency nodes and reinsert the parent. The final tree defines the codes.

EXAMPLE

For frequencies A:5, B:9, C:12, D:13, E:16, F:45, low-frequency symbols are merged first; frequent symbols usually get shorter codes.

COMPLEXITY

For k symbols, using a binary min-heap takes O(k log k) time and O(k) space.

TAKEAWAY

A prefix-free code means no symbol’s code is the prefix of another symbol’s code.